

 UNIVERSITY OF TECHNOLOGY VNUHCM FACULTY OF APPLIED SCIENCE	FINAL EXAM	Semester/Academic year: 232/2023-2024		
		Date: June 10		
	Course title	CALCULUS 2	Question sheet code: 3331	
	Course ID	MT1005		
	Duration	70 minutes		
- This is a closed book examination. Students are allowed to use standard pocket calculators. - You are required to fill in your full name and student ID on this paper.				
Student's full name: Student ID: Invigilator 1: Invigilator 2:				

- Numeric options (choices) in each question were rounded to the fourth decimal digit.
- There are 20 questions. Each correct answer will earn 0.5 points, while 0.1 points will be deducted for each incorrect answer. Leaving a question blank does not result in a deduction.
- The decimal point . is used when writing decimal numbers. For example: 0.5 (meaning $\frac{1}{2}$), or $1.0x$ (meaning x).
- In each question, select "None of the others" only if the other options are incorrect.

① (L.O.1) In $\mathbb{R}^3$, which equation from below gives an equation of a paraboloid?

- A. $z = x^2 + y^2$ D. None of the others
 B. $z = \sqrt{x^2 + y^2}$
 C. $z = -\sqrt{x^2 + y^2}$ E. $z = \sqrt{9 + x^2 + y^2}$

② (L.O.1) Let $f(x, y, z) = 0.7x^2y^3 + 0.9z^4 - 0.6$. Evaluate $f_x(0.7, 0.9, 0.6)$.

- A. None of the others C. 1.4144 E. 1.1144
 B. 0.7144 D. 1.2144

③ (L.O.2) Let $f(x, y)$ be a function such that $f(6, 4) = 0.3$, $f_x(6, 4) = -0.2$, $f_y(6, 4) = 1.6$. Use the linear approximation to obtain an approximation for $f(6.01, 3.98)$.

- A. 0.766 C. None of the others E. -0.034
 B. 0.466 D. 0.266

④ (L.O.1) Find a saddle point of the function $f(x, y) = 11(x - 8)^2 - 10(y - 5)^2$.

- A. (-8, -5) D. (8, 5)
 B. (11, 10)
 C. (0, 0) E. None of the others

⑤ (L.O.2) Find the maximum value of the function $f(x, y) = 1.3 - 2.3(x - 2.6)^2 - 1.9(y - 2.1)^4$ on the disk $x^2 + y^2 \leq 12.2$.

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| A. None of the others | C. 2.2 | E. 1.8 |
| B. 1.3 | D. 1.5 | |

6 (L.O.2) A lamina is bounded inside the triangle with vertices: $(0,0)$, $(1.4,0)$, $(0,1.4)$. The lamina has the density function $\rho(x,y) = 1$. Evaluate the mass of the lamina.

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| A. 1.28 | C. 0.68 | E. 0.98 |
| B. None of the others | D. 1.38 | |

7 (L.O.2) Evaluate

$$\iiint_E z dV,$$

where E is the solid bounded by the cylinder $x^2 + y^2 = 1.3$ and the planes: $z = 0$ and $z = 1$.

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| A. 1.642 | C. 2.242 | E. None of the others |
| B. 2.942 | D. 2.042 | |

8 (L.O.1) On the plane Oxy , which equation from the following gives the part of the circle of radius 10 centered at the origin that lies above the x -axis?

- A. None of the others
 B. $\langle 10 \cos(t), 10 \sin(t) \rangle$ for $t \in [-\pi/2, \pi/2]$
 C. $\langle 10 \cos(t), 10 \sin(t) \rangle$ for $t \in [0, \pi]$
 D. $\langle 10t, 10t^2 \rangle$ for $t \in [0, \pi]$
 E. $\langle 10 \cos(t), 10 \sin(t) \rangle$ for $t \in [0, 2\pi]$

9 (L.O.2) Evaluate the mass of a wire bent in the shape of the curve: $x = 2.0 \cos(t)$, $y = 2.0 \sin(t)$ for $t \in [0, 2\pi]$. Assume that the density of the wire is $\rho(x,y) = x^2$.

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| A. None of the others | C. 24.9327 | E. 26.0327 |
| B. 25.3327 | D. 25.1327 | |

10 (L.O.2) The base of a circular fence with radius 1.9 (m) is given by the curve: $x = 1.9 \cos(t)$, $y = 1.9 \sin(t)$ for $t \in [0, 2\pi]$. The height of the fence at position (x,y) is $h(x,y) = x^2 + y^2$ (m). Suppose that 1 (L) of paint covers 100 (m²). How much paint (in L) we will need if we paint both sides of the fence?

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| A. 1.2619 | C. 0.8619 | E. 1.1619 |
| B. None of the others | D. 0.4619 | |

11 (L.O.2) Find the work done by the force field $\mathbf{F}(x,y) = \langle -x^2, 2.0 \rangle$ in moving a particle along the curve $\langle 0.4 \cos(t), 0.4 \sin(t) \rangle$ for $0 \leq t \leq \pi$.

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| A. -0.2573 | C. None of the others | E. -0.3573 |
| B. -0.1573 | D. 0.0427 | |

- 12 (L.O.1) Evaluate the line integral

$$\oint_C xdx - yxdy,$$

where C is the boundary of the square with vertices: $(0, 0)$, $(3.3, 0)$, $(3.3, 3.3)$, and $(0, 3.3)$.

- A. None of the others C. -17.6685 E. -17.2685
B. -18.3685 D. -17.9685

- 13 (L.O.1) Let $\mathbf{n} = \langle a, b, c \rangle$ be the unit normal vector to the surface $z = 3 - 5x^2 - 4y^2$ at the point $M(1, -2, -18)$ in which $\mathbf{n}$ is pointing upward (namely $\widehat{\mathbf{n}, \vec{Oz}} < 90^\circ$). Find the value of c .

- A. 0.7529 C. 0.0529 E. None of the others
B. -0.1471 D. 0.3529

- 14 (L.O.2) Evaluate the surface area of the part of the plane $z = 2.6 + 1.3x$ that lies inside the cylinder $x^2 + y^2 = 1.3$.

- A. None of the others C. 6.6984 E. 7.0984
B. 6.9984 D. 6.3984

- 15 (L.O.1) Evaluate the surface integral $\iint_S \mathbf{F} \cdot d\mathbf{S}$ for the vector field $\mathbf{F}(x, y, z) = \langle xy, z, -x \rangle$, where S is the part of the plane $3.1x + y - z = 0$ that lies above the square: $0 \leq x \leq 1$, $0 \leq y \leq 1$, and has upward orientation.

- A. None of the others C. -2.625 E. -3.325
B. -2.825 D. -3.725

- 16 (L.O.1) Evaluate the surface integral $\iint_S \mathbf{F} \cdot d\mathbf{S}$ for the vector field $\mathbf{F}(x, y, z) = \langle y, z, -3.0zy^2 \rangle$, where S is the sphere $x^2 + y^2 + z^2 = 2.1$, and has outward orientation.

- A. -16.0616 C. -15.1616 E. None of the others
B. -15.5616 D. -15.7616

- 17 (L.O.1) Evaluate $\sum_{k=0}^{\infty} 2.7 \cdot (0.4)^k$.

- A. None of the others C. 4.5 E. 5.2
B. 5.4 D. 4.2

- 18 (L.O.1) Find the radius of convergence of the power series $\sum_{k=1}^{\infty} (2.0)^k (x - 1.3)^k$.

A. 0.5

C. 0.9

E. 1.2

B. None of the others

D. 0.1

19 (L.O.1) Which series from below is convergent?

A. None of the others

C. $\sum_{k=1}^{\infty} 2.0$

E. $\sum_{k=1}^{\infty} 2^k$

B. $\sum_{k=1}^{\infty} \frac{3.4}{k^2}$

D. $\sum_{k=1}^{\infty} \frac{3.4}{k}$

20 (L.O.1) Find a value of x from the following in which the series $\sum_{k=1}^{\infty} \frac{1}{k} (x - 1.9)^k$ is convergent.

A. 2.2

C. 5.2

E. 6.2

B. 4.2

D. None of the others

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1. (A)
2. (B)
3. (D)
4. (D)
5. (B)
6. (E)
7. (D)
8. (C)
9. (D)
10. (C)
11. (D)
12. (D)
13. (C)
14. (C)
15. (E)
16. (A)
17. (C)
18. (A)
19. (B)
20. (A)